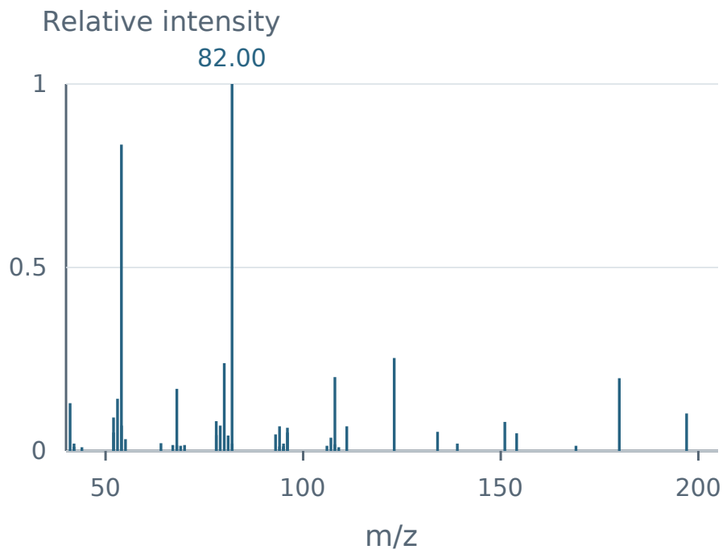
Identify a molecule from its fragmentation spectrum

Different molecular structures can share nearly the same mass. MS/MS fragments help distinguish them.

A Measured spectrum

Precursor m/z 197.0305 · [M+H]+

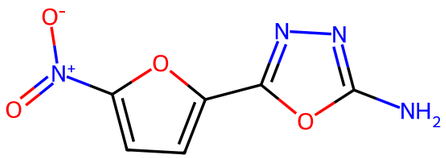


45 measured peaks · QTOF · validation

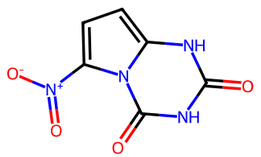
B Candidate structures

3 of 256 mass-compatible candidates

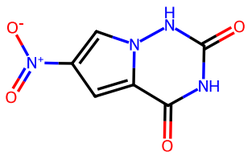
A



B



C



Candidate IDs label structures, not model scores.

C Retrieval output

Actual archived GLACIER validation ranking

Rank candidates using fragment evidence

Spectrum + metadata + candidate molecules

RANK	CANDIDATE
1	B
2	C
22	A — validation target

True structure ranks 22 / 256: a retrieval error.

One example; not an estimate of overall performance.

Scientific value

Accurate molecular identification supports metabolomics, chemical analysis, and the discovery of unknown small molecules.

Benchmark scope: rank supplied, mass-compatible structures from MS/MS data; the primary metric is exact-candidate Hit@1.

Example: MassSpecGymID0079471 (validation). Source: MassSpecGym developers, 2024, MIT. Molecules and ranking are real; see example.json for provenance.